Introduction To Lid Disease

Lorne Yudcovitch, OD, MS, FAAO

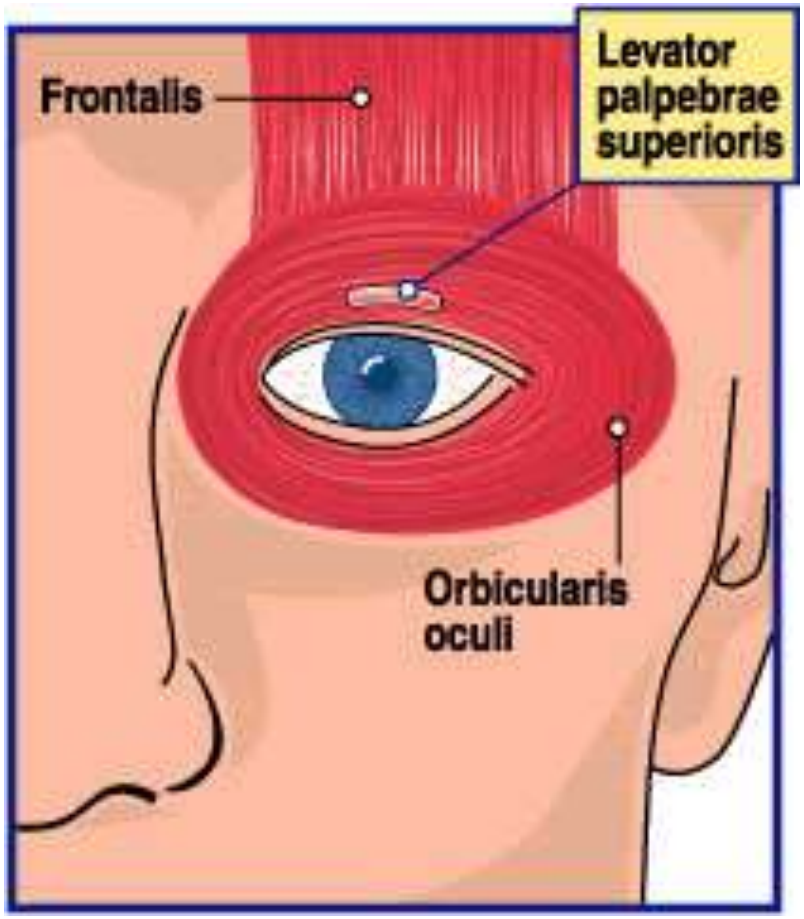
SOAP Format

- **S = Subjective**
  - **AKA symptoms**
  - **Everything the patient says**
- **O = Objective**
  - **AKA signs**
  - **All the doctors observes and all the test results**
- **A = Assessment**
  - **AKA diagnosis (or diagnoses)**
  - **What is wrong with the patient**
- **P = Plan**
  - **For each diagnosis, what is going to be done to help the patient**

Lid Anatomy

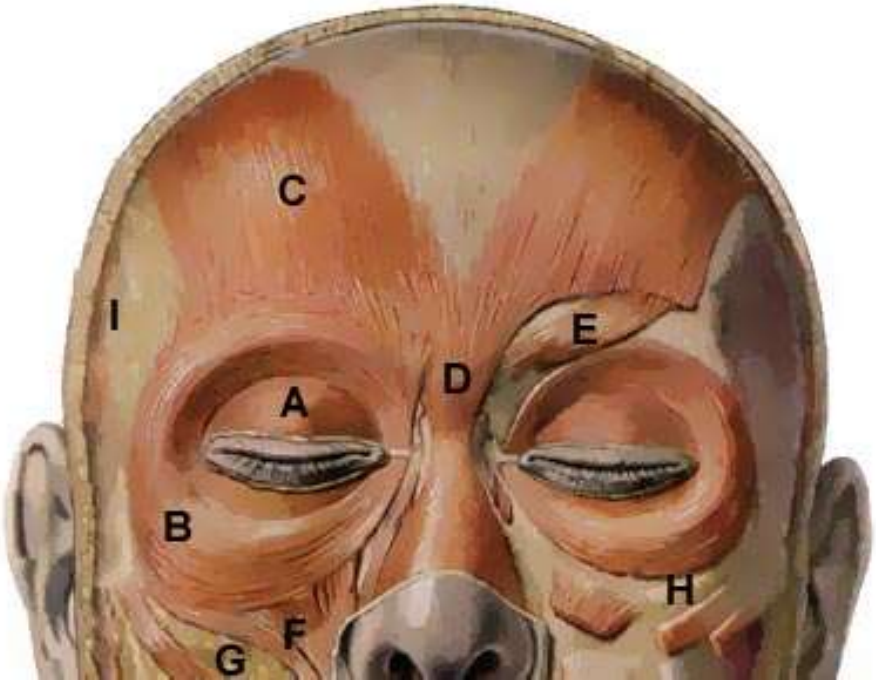
Muscles

- Levator
- Mueller's muscle
- Orbicularis



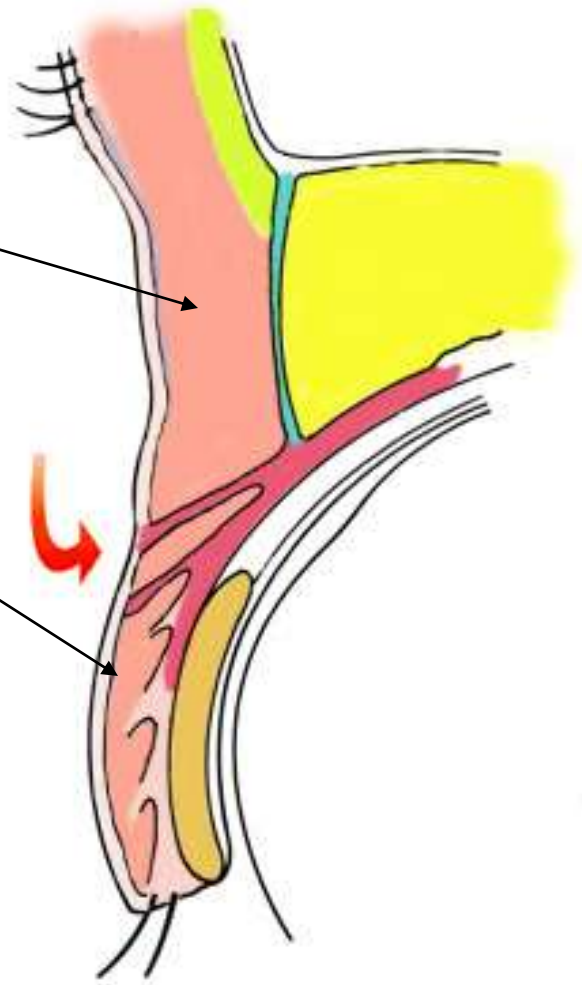
Lid Anatomy

- **A = orbicularis muscle (palpebral portion)**
  - **Striated**
  - **Eyelid closing muscle - used in blinking and voluntary winking**
  - **Keeps lid tight to globe**
- **B = orbicularis muscle (orbital portion)**
  - **Striated**
  - **Eyelid closing muscle - used in forced closure**



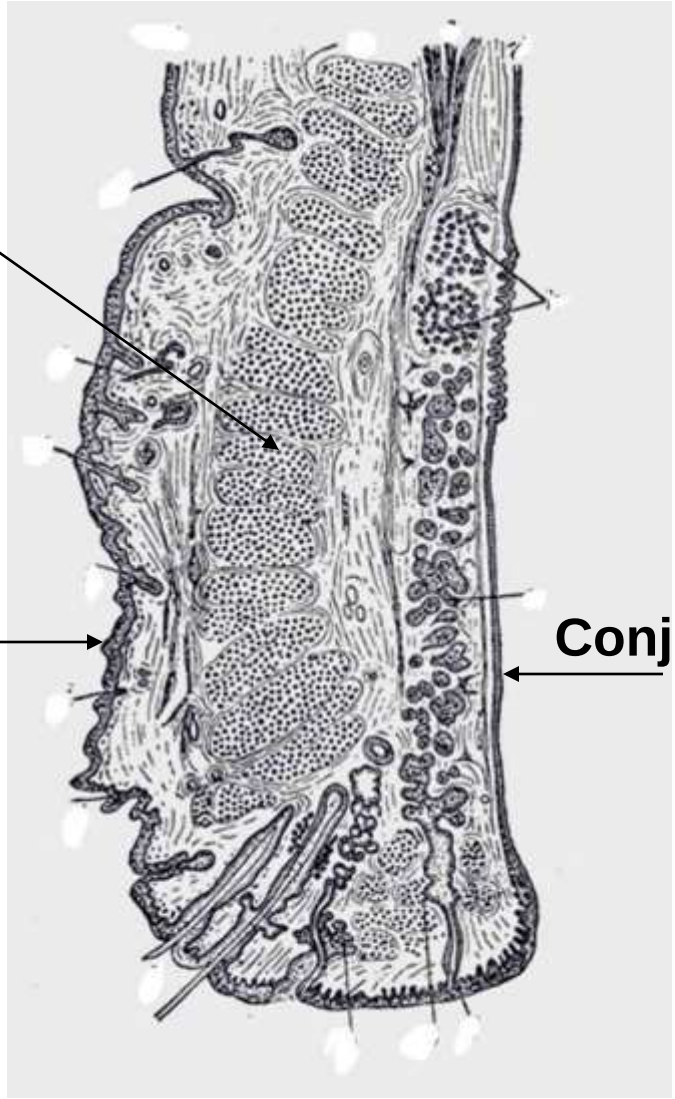
Lid Anatomy

- Orbicularis - orbital portion
- Orbicularis - palpebral portion



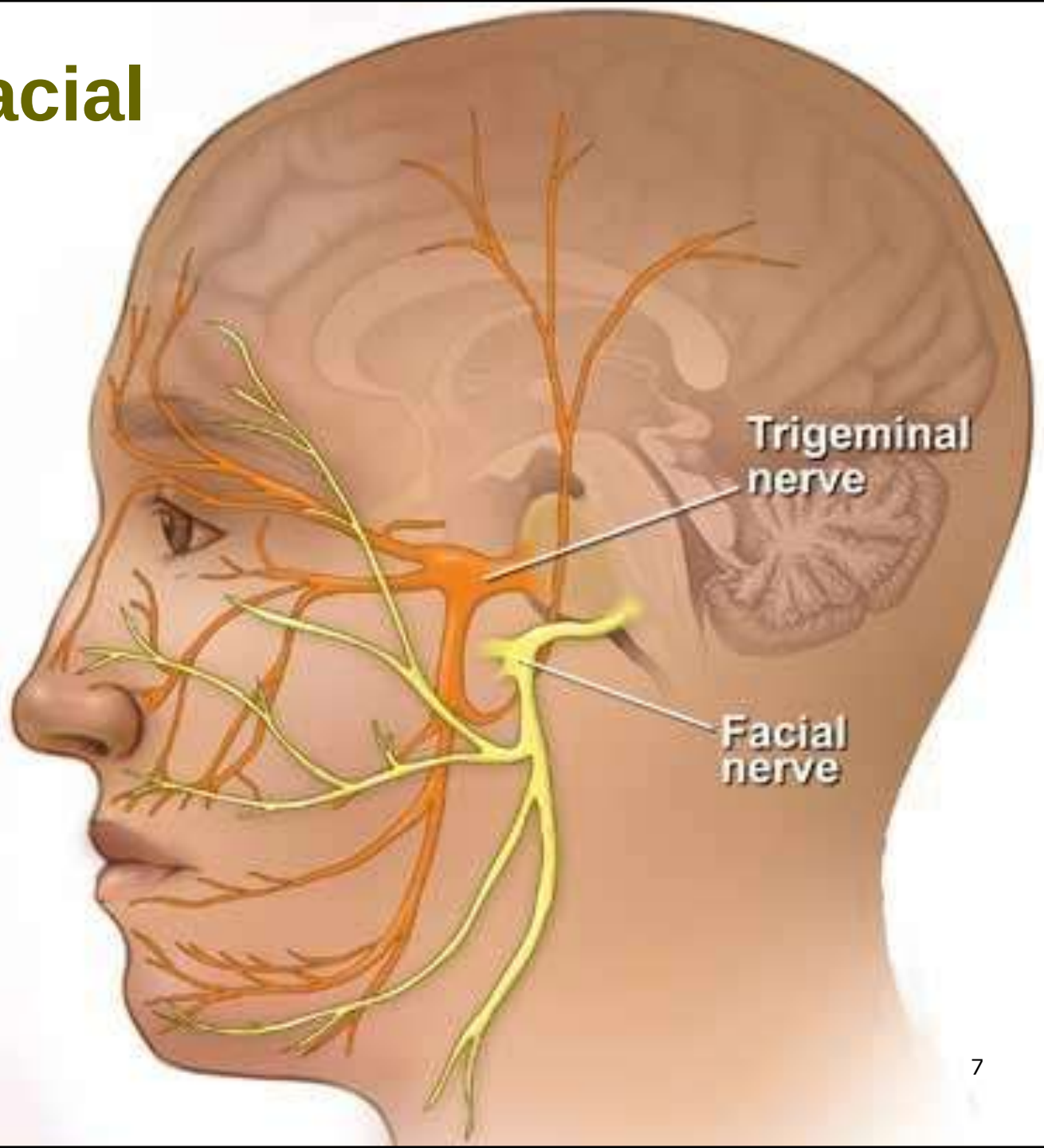
Lid Anatomy

- **Orbicularis**
  - **Innervation is from the facial nerve (CN VII)**

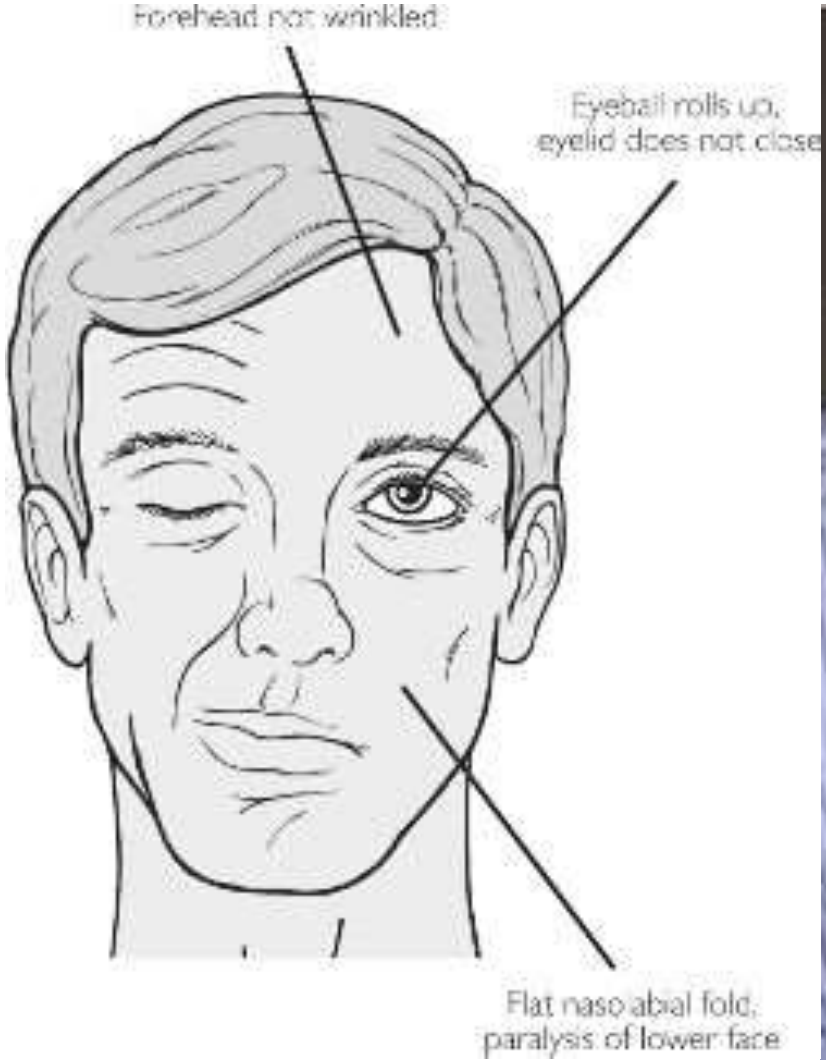


CNVII = Facial Nerve

- CNVII causes:
- eyelids to close
- mouth and cheeks to ‘smile’
- forehead to ‘furrow’



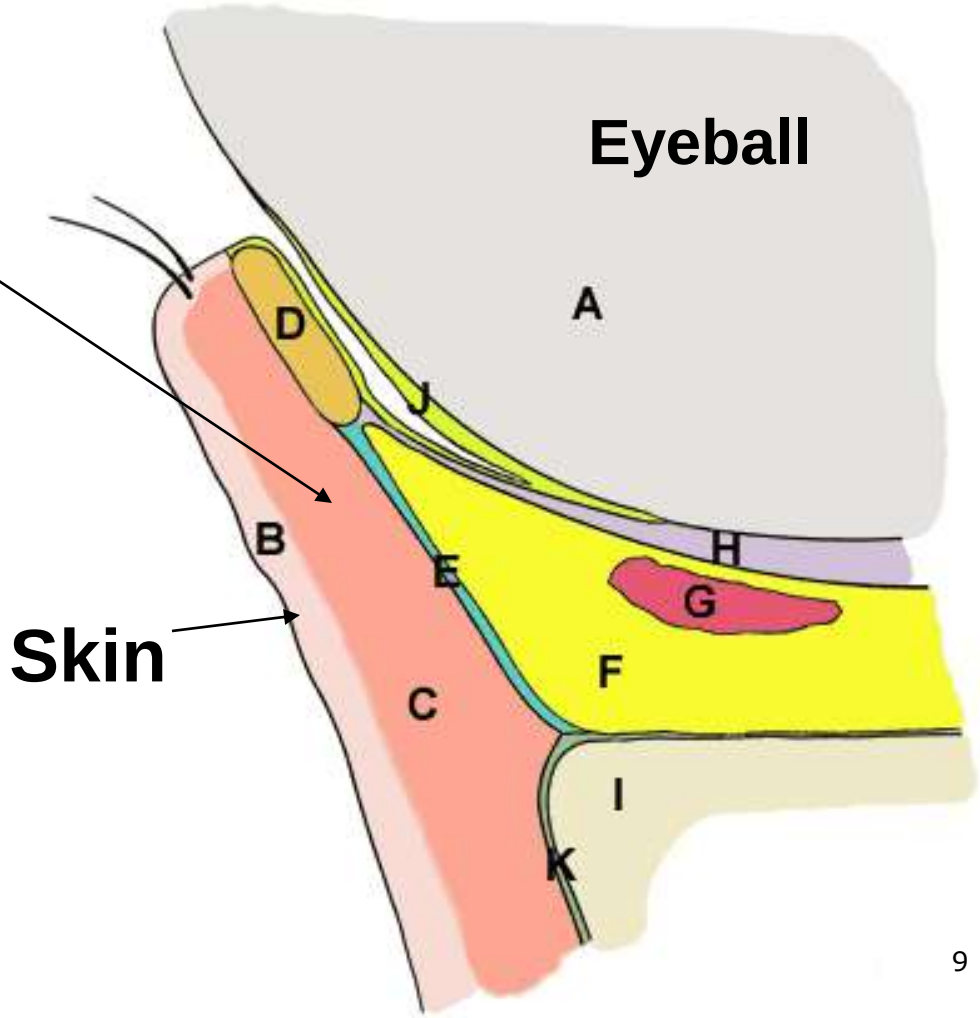
Bell's Palsy


Forehead not wrinkled
Eyeball rolls up, eyelid does not close
Flat naso-abial fold, paralysis of lower face



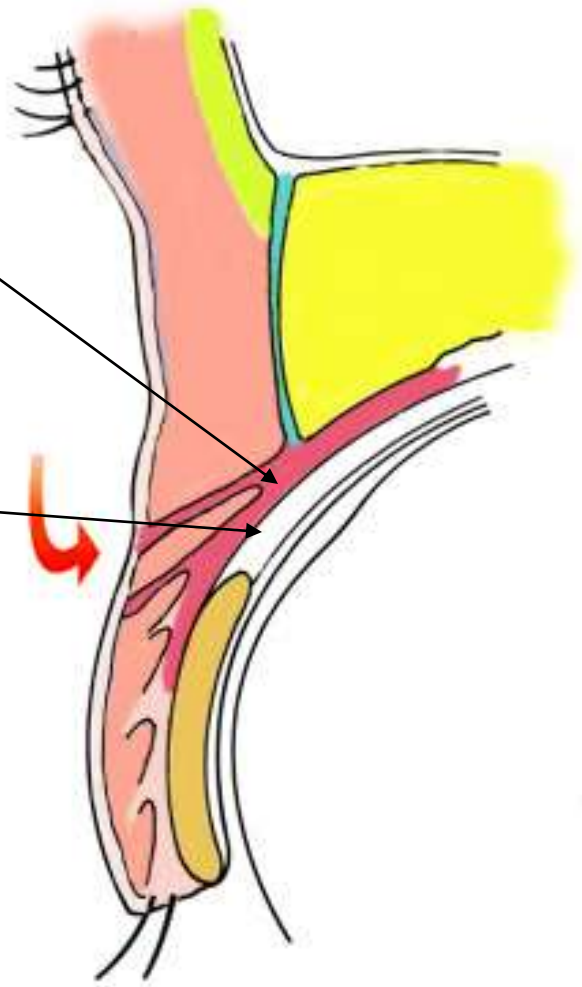
Lid Anatomy

- **Orbicularis (lower lid)**



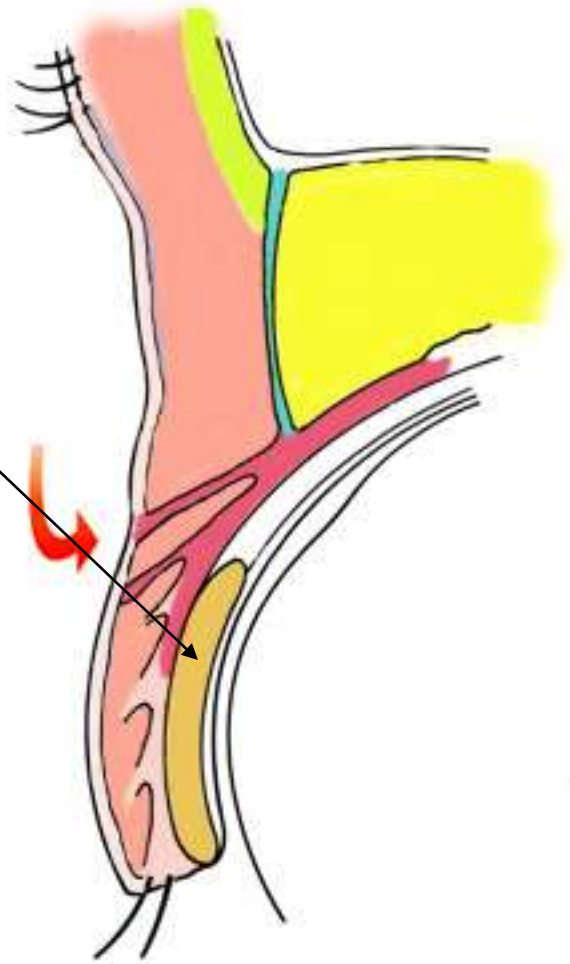
Lid Anatomy

- **Levator aponeurosis**
  - Striated
  - Lid opening muscle
  - Innervated by CN III
  - Fibers pass through the orbicularis to attach to the skin
- **Muller's muscle**
  - Originates from the underside of the levator
  - Smooth
  - Lid opening muscle
  - Innervated by the sympathetic system
  - Inserts into superior tarsus



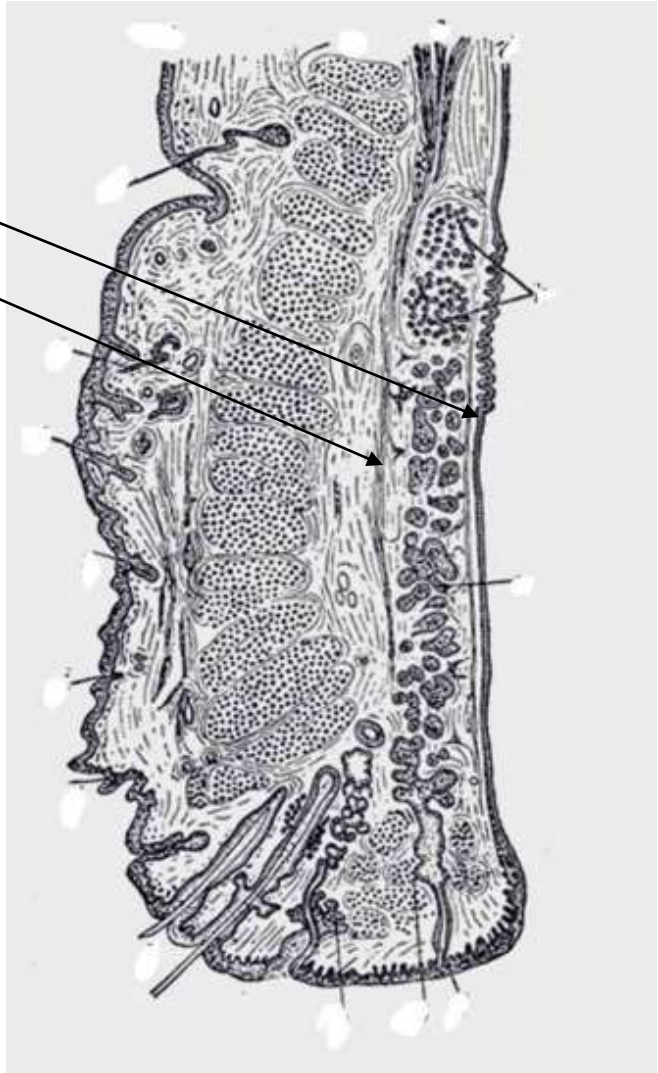
Lid Anatomy

- **Tarsal plate**
  - **Dense fibrous connective tissue**
  - **Responsible for structural integrity of the lids**
  - **Upper plate about 1 cm high x 2.5 cm wide**
  - **Lower tarsal plate smaller dimensions**



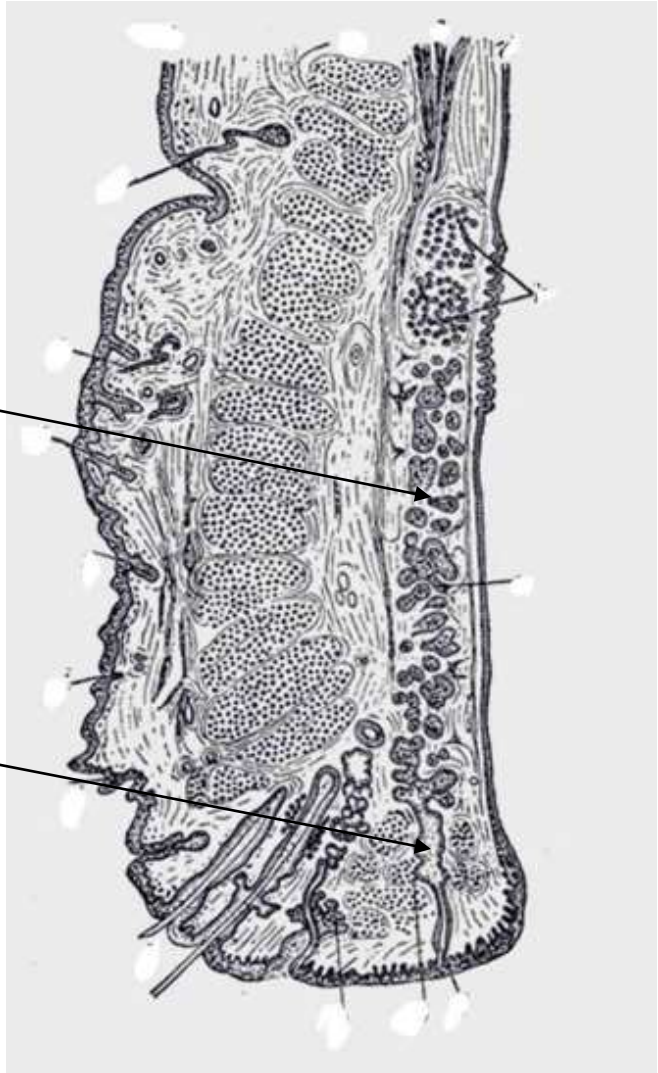
Lid Anatomy

- **Tarsal plate (between arrows)**



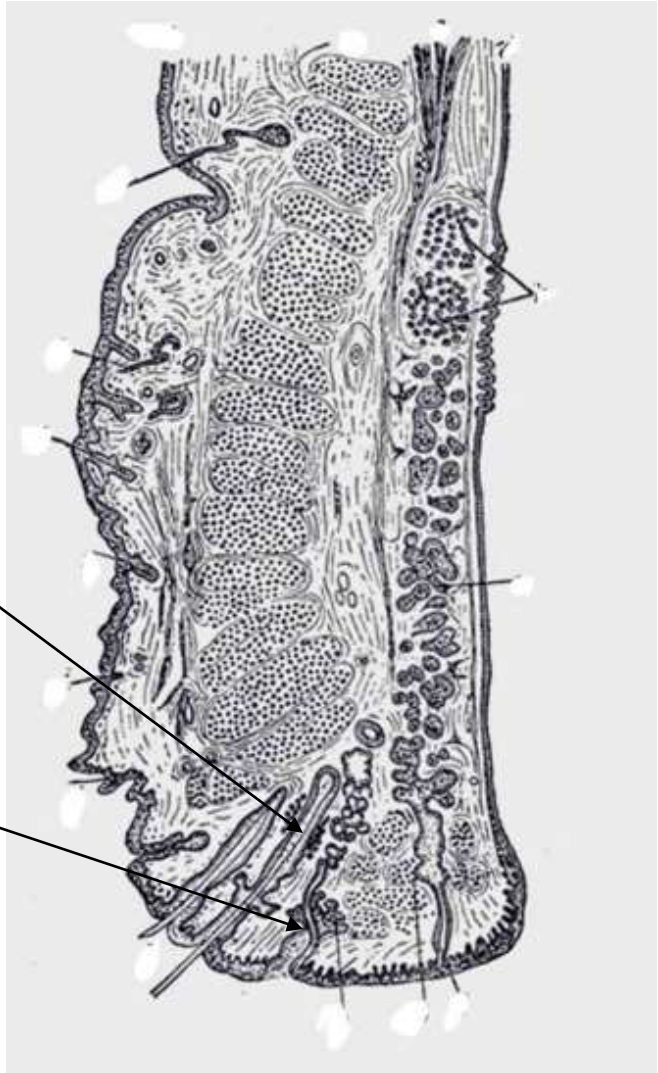
Lid Anatomy

- Each tarsus encloses about 25 meibomian glands
- Meibomian gland
  - Sebaceous gland (oil)
  - Provides the oily layer of the tear film
  - Meibomian gland opening just behind the lashes



Lid Anatomy

- **Lash follicle and associated sebaceous gland(s) (Zeiss gland)**
- **Moll gland - sweat**



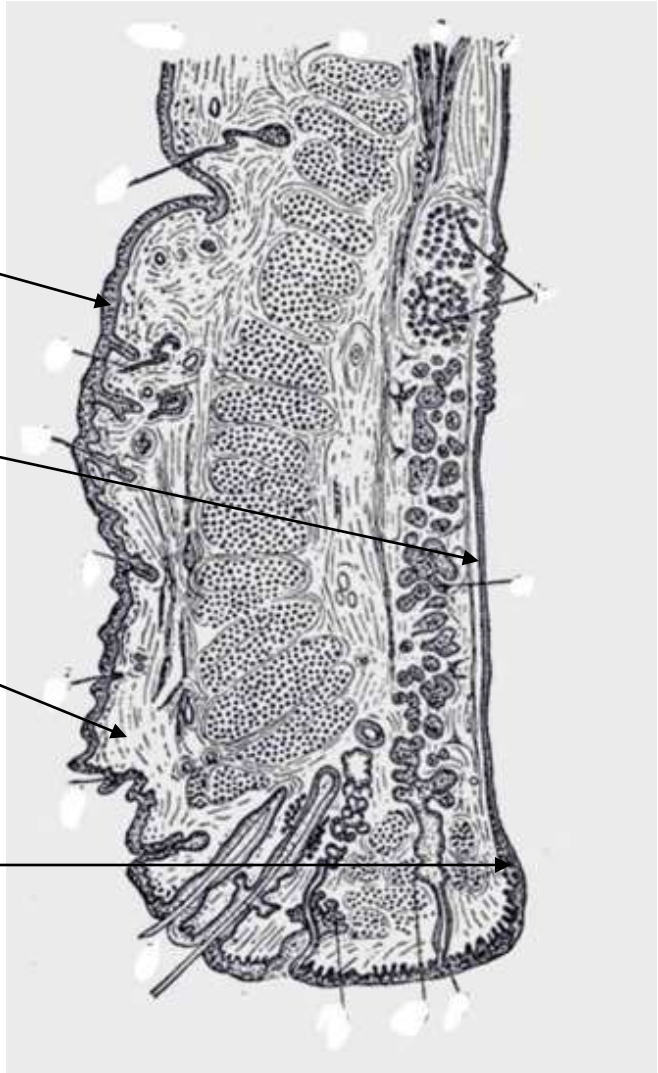
Lid Anatomy

Skin

Conjunctiva

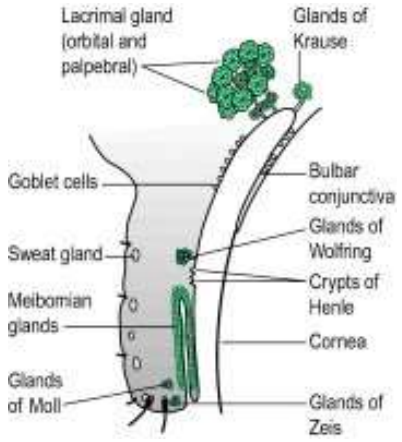
Loose connective tissue

Mucocutaneous junction



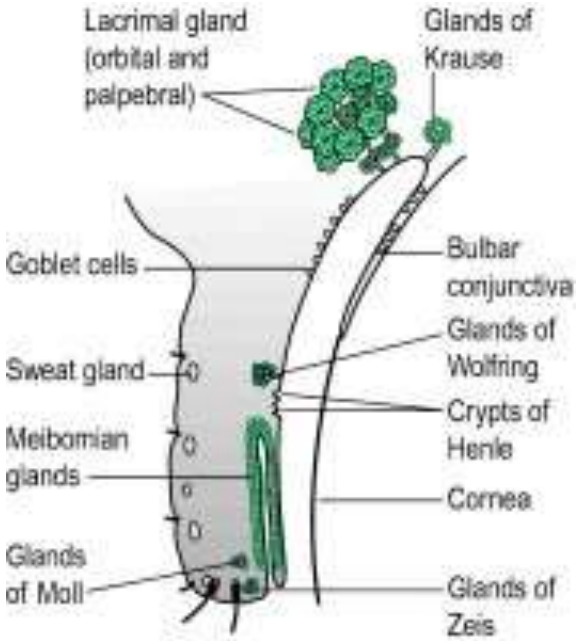
Eyelid Glands – Conjunctival Side

- **Goblet cells**
  - In conjunctival epithelium
  - Secrete mucin
  - Abundant at plica semilunaris (caruncle)
  - None on limbus or mucocutaneous junction
- **Accessory glands of Krause and Wolfring**
  - Krause – near fornix; basal aqueous secretion
  - Wolfring – near tarsus; basal aqueous secretion
  - Both assist lacrimal gland in aqueous secretion
  - Only lacrimal gland has reflex secretion (tearing)



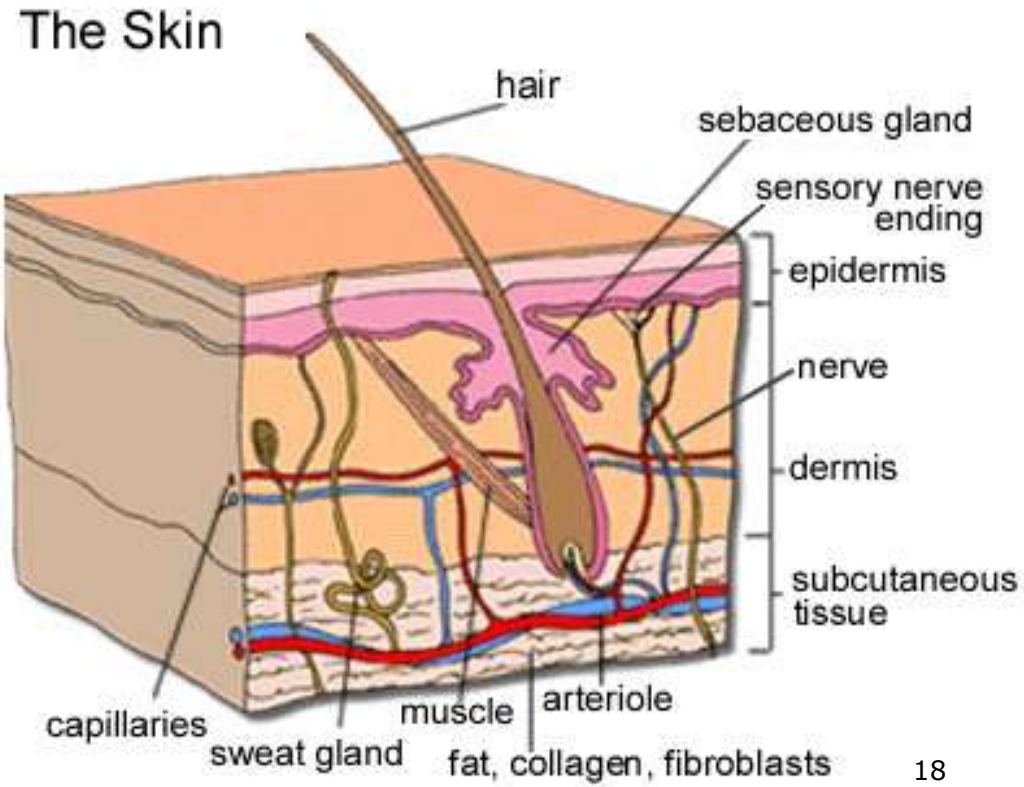
Eyelid Glands – Lid Margin and Skin

- **Meibomian glands**
  - **Secrete sebum (lipid)**
- **Gland of Zeiss**
  - **Unilobular gland**
  - **Produce oils for the lash follicles**
- **Gland of Moll**
  - **aka ciliary gland**
  - **Anterior to Zeiss glands**
  - **Produce oils as well**
- **Sweat gland**
  - **aka sudoriferous gland**

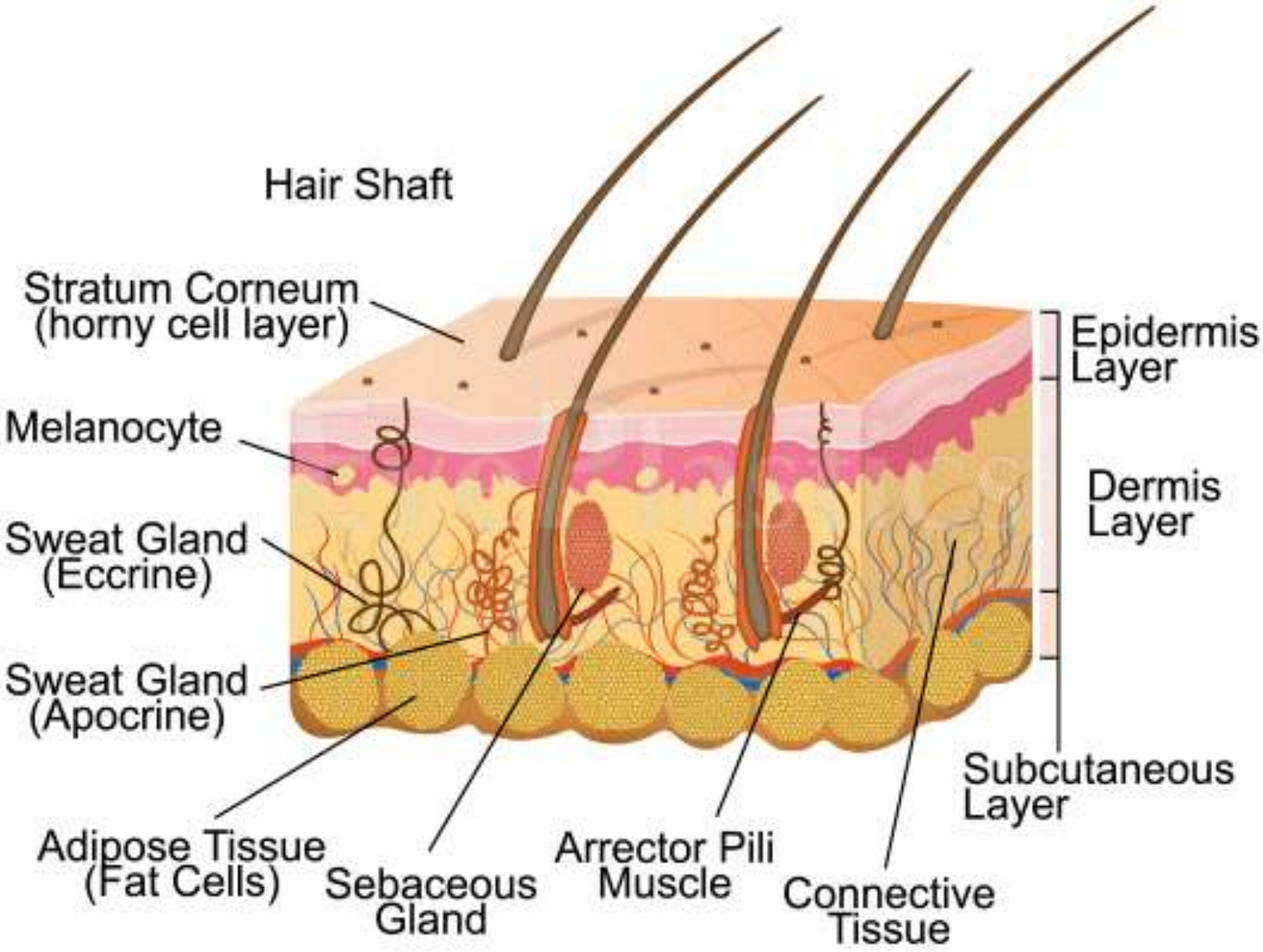


Lid Anatomy - Skin

- **Eyelid skin is the thinnest of the body**
  - **Epidermis**
  - **Dermis**
  - **Hair follicle and associated sebaceous gland**
  - **Sweat gland**
  - **Blood supply**

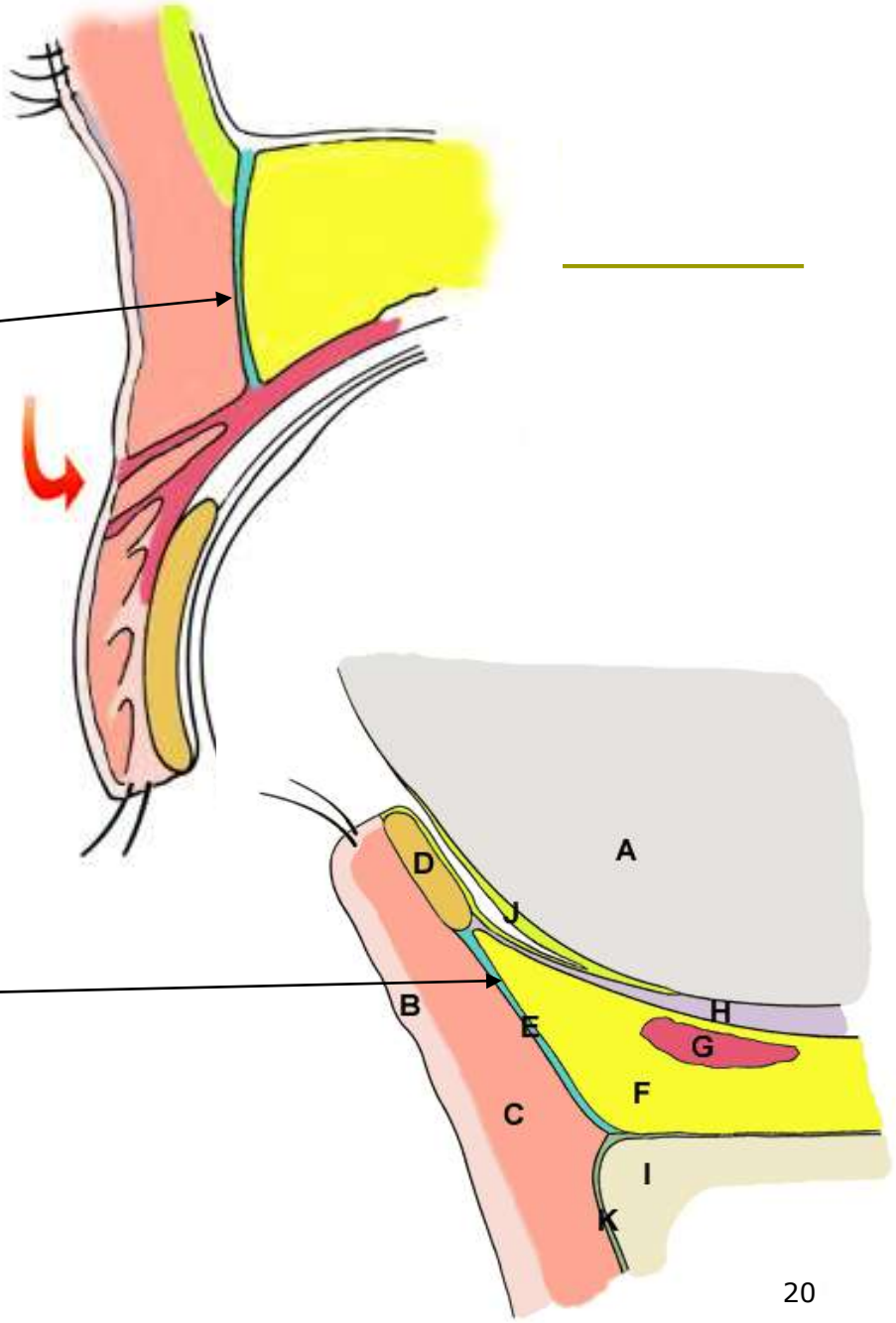


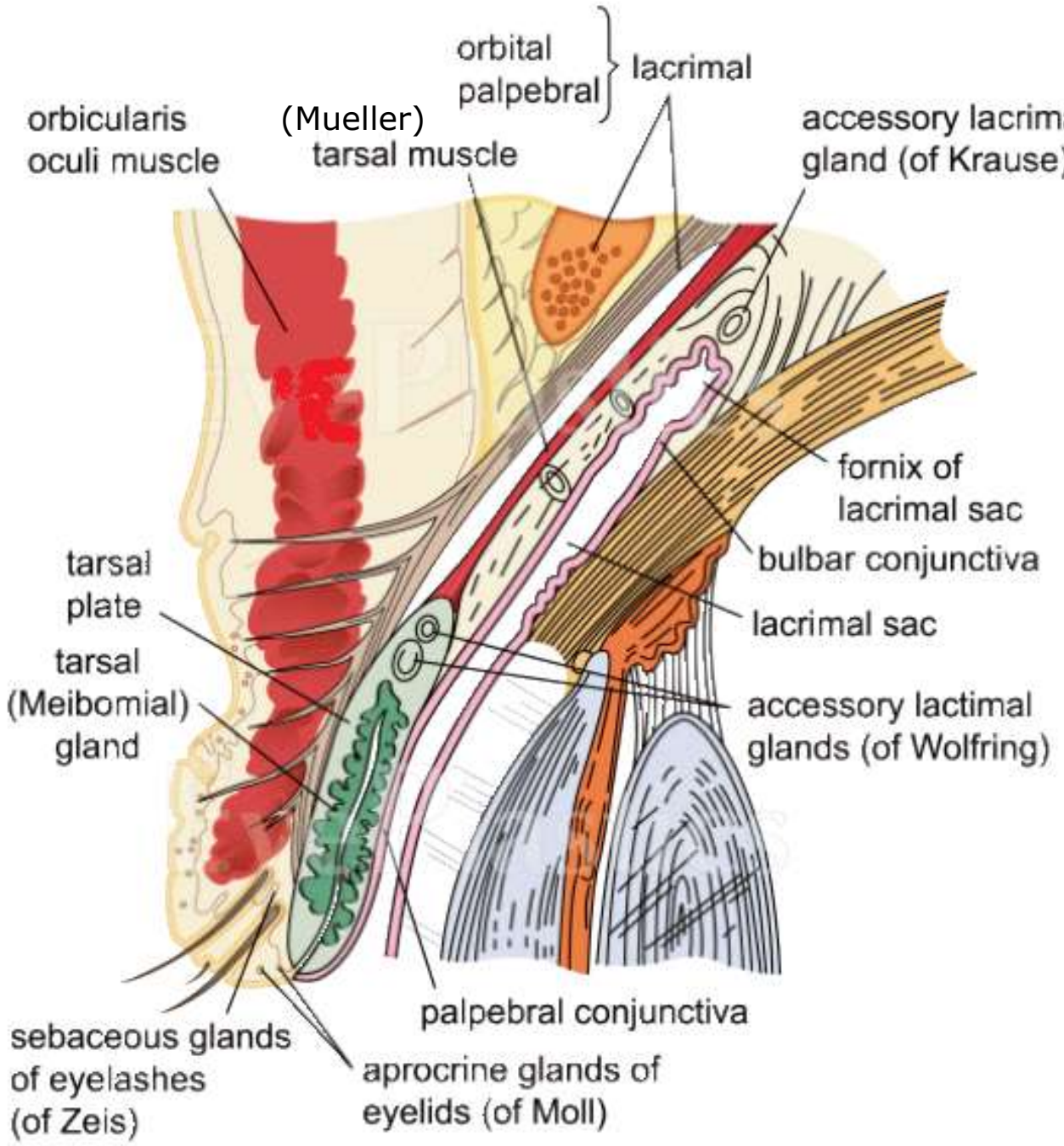
Skin Anatomy



Lid Anatomy

- **Orbital septum**
  - **Attaches peripherally at the periosteum at the orbital margin**
  - **Centrally, it fuses with levator aponeurosis (upper lid) and the tarsal plate (lower lid)**
  - **Acts as diaphragm to separate orbital structures from lid structures**





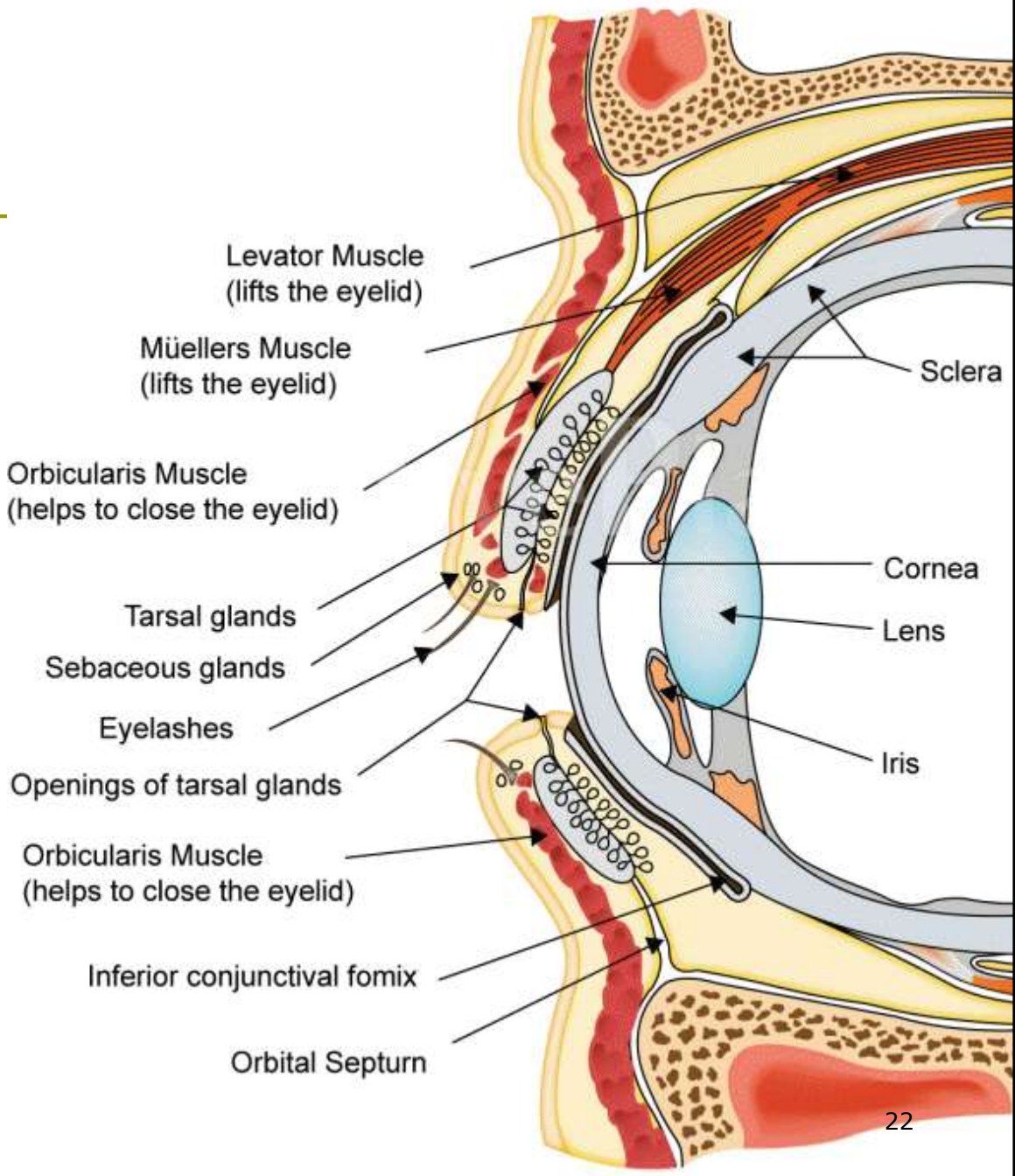
Eyelids are complex!

- Skin
- Hair
- Fascia
- Muscle
  - Striated
  - Smooth
- Glands
- Nerves
- Conjunctiva
- Connective tissue

…lots of structures for disease to occur!

Lid Diseases

- Position/shape
- Inflammation
- Allergy
- Infection
- Cancer
- Autoimmune
- Gland dysfunction
- Neurological
- Trauma



THANK YOU!